KUNNETH FORMULA FOR DERIVED PUSHFORWARDS ALONG TOR-INDEPENDENT QCQS MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-cohomology-base-change and lemma-compare-base-change are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 5666 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Cohomology and base change along the factors. lemma-cohomology-base-change (same chapter, NOT carded, declared external). Two of the four displayed equalities (Rp_* of a pullback and the final tensor identification).
- Base-change comparison morphism. lemma-compare-base-change (same chapter, NOT carded, declared external). Identifies Rp_* Lq^* M with La^* Rb_* M under tor-independence.
- Projection formula for derived pushforwards. Cohomology, lemma projection formula (cross-chapter, named). Used twice to move the tensor past Ra_* and Rp_*.

Source reference. Original equation 0FLP: sources/stacks-perfect/pinned-original-source.tex, source lines 5610–5610. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0G7V: sources/stacks-perfect/pinned-original-source.tex, source lines 5623–5623. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation above, if a and b are quasi-compact and quasi-separated and X and Y are tor-independent over S, then (0FLP) is an isomorphism for $K \in D_{QCoh}(\mathcal{O}_X)$ and $M \in D_{QCoh}(\mathcal{O}_Y)$. If in addition $S = \mathrm{Spec}(A)$ is affine, then the map (0G7V) is an isomorphism.*

First proof. This follows from the following sequence of isomorphisms

$$\begin{aligned} Rf_*(Lp^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} Lq^*M) &= Ra_*Rp_*(Lp^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times_S Y}} Lq^*M) \\ &= Ra_*(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} Rp_*Lq^*M) \\ &= Ra_*(K \otimes^{\mathbf{L}}_{\mathcal{O}_X} La^*Rb_*M) \\ &= Ra_*K \otimes^{\mathbf{L}}_{\mathcal{O}_S} Rb_*M \end{aligned}$$

The first equality holds because $f = a \circ p$. The second equality by Lemma 08EU. The third equality by Lemma 08IB. The fourth equality by Lemma 08EU. We omit the verification that the composition of these isomorphisms is the same as the map (0FLP). If S is affine, then the source and target of the arrow (0G7V) are the result of applying $R\Gamma(S, -)$ to the source and target of (0FLP) and we obtain the final statement; details omitted. □